

Business Requirement Document (BRD)

IntelliPlant AI

AI-Powered Industrial Knowledge Intelligence Platform

Document Version: 1.0

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1. Document Purpose

This document defines the business objectives, challenges, stakeholders, requirements, benefits, constraints, and success criteria for IntelliPlant AI.

The goal is to create a unified AI-powered knowledge intelligence platform that transforms fragmented industrial information into a searchable, explainable, and continuously evolving knowledge system.

2. Executive Summary

Industrial organizations maintain large volumes of documents distributed across multiple systems, including maintenance records, SOPs, engineering drawings, inspection reports, manuals, and email archives.

This fragmented information environment causes:

- Long information retrieval times
- Knowledge loss
- Repeated failures
- Compliance issues
- Increased downtime
- Reduced productivity

IntelliPlant AI will establish a centralized intelligence layer that combines:

- Document Intelligence
- Retrieval-Augmented Generation (RAG)
- Knowledge Graphs
- Multi-Agent AI
- Root Cause Analysis

- Compliance Intelligence
- Lessons Learned Engine

The platform aims to provide engineers and managers with accurate and explainable answers supported by source documents and graph relationships.

3. Business Problem

Current Situation

Industrial organizations typically operate with:

- 7-12 disconnected systems
- Thousands of documents
- Scattered maintenance records
- Email archives
- Engineering drawings
- Inspection reports

Information exists but cannot be easily accessed.

Problems Faced

Knowledge Silos

Information is distributed across departments.

Knowledge Loss

Expert knowledge disappears when experienced engineers retire.

Long Search Time

Engineers spend significant time searching for information.

Repeated Failures

Lessons learned from past incidents are not reused.

Compliance Risks

Manual audit preparation increases risk.

Operational Downtime

Missing historical information causes delayed troubleshooting.

4. Business Opportunity

By creating an Industrial Knowledge Intelligence Platform, organizations can:

- Reduce downtime
 - Increase productivity
 - Improve compliance
 - Capture tribal knowledge
 - Improve reliability
 - Accelerate decision making
 - Improve cross-functional collaboration
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5. Business Objectives

BO-01

Reduce information retrieval time by 80%.

BO-02

Preserve organizational knowledge.

BO-03

Improve maintenance efficiency.

BO-04

Enable intelligent root cause analysis.

BO-05

Improve audit readiness.

BO-06

Prevent repeated incidents.

BO-07

Increase operational reliability.

BO-08

Improve cross-functional collaboration.

6. Stakeholders

Executive Management

Responsibilities:

- Strategic decisions
- Asset performance

Needs:

- Insights
 - Risk reports
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Plant Managers

Responsibilities:

- Operations
- Asset availability

Needs:

- Operational intelligence
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Maintenance Team

Responsibilities:

- Repairs
- Preventive maintenance

Needs:

- Equipment history
 - Similar failures
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Reliability Engineers

Responsibilities:

- Root cause analysis
- Reliability improvement

Needs:

- Failure intelligence
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Quality Team

Responsibilities:

- Compliance
- Audits

Needs:

- Evidence generation
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Safety Team

Responsibilities:

- Incident management

Needs:

- Lessons learned intelligence
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Field Technicians

Responsibilities:

- Equipment inspection

Needs:

- Mobile guidance
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7. Target Industries

Oil and Gas

Refineries

Petrochemical Plants

Pipelines

Power Generation

Thermal Plants

Hydroelectric Plants

Renewable Energy Facilities

Manufacturing

Steel Plants

Automobile Plants

Food Processing

Chemical Plants

Mining

Coal

Metal Processing

Cement Industry

EPC Organizations

Construction

Data Centers

Infrastructure Projects

8. Expected Business Benefits

Reduced Downtime

Faster issue diagnosis.

Higher Productivity

Less time spent searching documents.

Knowledge Preservation

Expert knowledge remains accessible.

Better Decision Making

AI-assisted recommendations.

Improved Compliance

Automatic evidence collection.

Reduced Failure Recurrence

Historical lessons become reusable.

Improved Collaboration

Unified information sharing.

9. Business Capabilities

The system should provide:

Universal Document Intelligence

Support:

- PDF
 - DOCX
 - Excel
 - Images
 - Emails
 - CAD Drawings
 - P&ID Documents
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Knowledge Discovery

Find relationships between:

- Equipment
- Failures
- Procedures

- Personnel
 - Regulations
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Expert Copilot

Conversational AI assistant.

Root Cause Analysis

Identify probable causes.

Compliance Intelligence

Detect gaps against:

- Factory Act
 - OISD
 - PESO
 - ISO Standards
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Lessons Learned Engine

Identify recurring patterns.

Maintenance Intelligence

Recommend actions.

10. Business Requirements

BR-01

Users shall upload heterogeneous documents.

BR-02

The platform shall perform OCR and document understanding.

BR-03

The platform shall create a knowledge graph.

BR-04

Users shall search documents using natural language.

BR-05

Answers shall include citations.

BR-06

The system shall provide confidence scores.

BR-07

The system shall support multi-agent reasoning.

BR-08

The system shall preserve historical incidents.

BR-09

The system shall detect compliance gaps.

BR-10

The system shall generate lessons learned.

11. Assumptions

Documents are available in digital form.

Users possess domain knowledge.

Industrial standards remain stable.

Internet connectivity exists.

12. Constraints

Limited availability of structured datasets.

Domain-specific terminology.

Large document sizes.

Complex engineering drawings.

Confidential information handling.

13. Risks

Knowledge Extraction Errors

Mitigation:

Human validation.

Hallucination Risk

Mitigation:

RAG with citations.

Missing Relationships

Mitigation:

Knowledge graph enrichment.

Incomplete Documents

Mitigation:

Incremental ingestion.

Regulatory Changes

Mitigation:

Continuous updates.

14. Key Performance Indicators

KPI	Target
Information Retrieval Time	<20 sec
Search Accuracy	>90%
Citation Accuracy	>95%
User Satisfaction	>85%
Compliance Gap Detection Accuracy	>90%
Knowledge Graph Coverage	>85%
Response Latency	<5 sec
Hallucination Rate	<5%

15. Success Criteria

The project shall be considered successful if:

- Engineers retrieve answers within seconds.
- Source citations are available.
- Knowledge graph relationships are created automatically.
- Compliance gaps are identified.
- Historical incidents are reusable.
- Root cause analysis becomes faster.
- Cross-functional knowledge sharing improves.

16. Future Vision

IntelliPlant AI evolves into a complete Industrial Operations Brain supporting:

- Digital Twins
- IoT Integration
- Predictive Maintenance
- SCADA Integration
- Autonomous Agents
- Real-Time Asset Intelligence
- Industrial Copilots

Conclusion

IntelliPlant AI addresses one of the largest challenges in industrial operations—knowledge fragmentation. By combining document intelligence, knowledge graphs, and multi-agent AI, the platform transforms static information into actionable organizational intelligence, improving reliability, safety, compliance, and operational excellence.